

3. To define the associator, notice that for $V, W, U \in \text{Rep}(A)$, both the spaces $(V \overline{\otimes} W) \overline{\otimes} U$ and $V \overline{\otimes} (W \overline{\otimes} U)$ are given by the retract of the idempotent

$$V \otimes W \otimes U \longrightarrow V \otimes W \otimes U, \quad v \otimes w \otimes u \longmapsto 1_{(1)} \cdot v \otimes 1_{(2)} \cdot w \otimes 1_{(3)} \cdot u,$$

which can be verified using **(Axiom 1)**. We then define the associator $a_{V,W,U}$ as the canonical map between the two retracts. One can check that $a_{V,W,U}$ is an A -module map, and satisfies the pentagon equation.

4. Given $V \in \text{Rep}(A)$, we set the left unitor $l_V: A^l \overline{\otimes} V \longrightarrow V$ to be the restriction of the map

$$A^l \otimes V \longrightarrow V, \quad x \otimes v \longmapsto x.v$$

on $A^l \overline{\otimes} V$; we set the right unitor $r_V: V \overline{\otimes} A^l \longrightarrow V$ to be the restriction of the map

$$V \otimes A^l \longrightarrow V, \quad v \otimes y \longmapsto \varepsilon^{rr}(y).v$$

on $V \overline{\otimes} A^l$. One can check that l_V and r_V are indeed invertible A -module maps, and satisfy the triangle equations.

5. The left dual V^L of an object $V \in \text{Rep}(A)$ is given by the dual vector space $V^* := \text{Hom}(V, k)$ endowed with the A -action

$$x.\omega = \omega(S(x).-), \quad \forall \omega \in V^*, x \in A.$$

Similarly, the right dual V^R of V is given by V^* endowed with A -action

$$x.\omega = \omega(S^{-1}(x).-), \quad \forall \omega \in V^*, x \in A.$$

Secondly, note that $\text{Rep}(A)$ is clearly a finite k -linear category, with the tensor product $\overline{\otimes}$ being bi- k -linear. This concludes our construction of $\text{Rep}(A)$.

1.20 Example. We illustrate the above construction of $\text{Rep}(A)$ when A is the weak Hopf algebra $B \otimes B^{\text{op}}$ defined in Example 1.17. Since a left A -module is precisely a B - B -bimodule, $\text{Rep}(A)$ is equivalent to $\text{BiMod}(B|B)$ as categories. It remains to find the monoidal structure on $\text{Rep}(A)$. Given left A -modules V and W , which we identify as B - B -bimodules, the underlying vector space of $V \overline{\otimes} W$ is the retract of the idempotent

$$V \otimes W \longrightarrow V \otimes W, \quad v \otimes w \longmapsto v.p^{(1)} \otimes p^{(2)}.w.$$

The action of $a \otimes b \in B \otimes B^{\text{op}}$ on $V \overline{\otimes} W$ is given by the restriction of the map

$$V \otimes W \longrightarrow V \otimes W, \quad v \otimes w \longmapsto a.v.p^{(1)} \otimes p^{(2)}.w.b$$

on $V \overline{\otimes} W$. Using Corollary 1.5, it can be shown that the B - B -bimodule $V \overline{\otimes} W$ is precisely $V \otimes_B W$.

To find the tensor unit of $\text{Rep}(B \otimes B^{\text{op}})$, one first computes

$$\varepsilon^{lr}: B \otimes B^{\text{op}} \longrightarrow B \otimes B^{\text{op}}, \quad a \otimes b \longmapsto ab \otimes 1;$$

$$\varepsilon^{rr}: B \otimes B^{\text{op}} \longrightarrow B \otimes B^{\text{op}}, \quad a \otimes b \longmapsto 1 \otimes ab.$$